

How Do You Know It Worked?

The instruments, the illusions, and the surrogate endpoints behind every claim about oxygen delivery

Suppose a patient in shock receives a BHOC and, an hour later, looks better. The blood pressure has come up. The skin is warmer. The lactate is falling. Did the oxygen carrier do that?

The honest answer depends almost entirely on what you measured, how, and with what instrument, and the instruments that we rely on were built for a world without free hemoglobin in plasma. This post walks through the measuring tools of oxygen delivery, what each can tell you, and the ways each can mislead when a cell-free carrier is present. It may be the most technical post on this site, but it is also the one that most affects whether a claim can be believed.

The problem in one sentence

Oxygen delivery is the product of blood flow and the oxygen content of blood, and the aim of delivery is not the blood but the tissue. Everything measured at the bedside is an indirect reading of what the cells are actually receiving.

A carrier can raise blood oxygen content and still fail the tissue, if it changes flow. It can look useless on a global measure and still rescue a small region. And it can distort the test itself. Each of the following tools has a way of being wrong.

Hemoglobin concentration: the number that stops meaning what it meant

The most common lab value in medicine is the hemoglobin. Every clinician reads it as a count of red cell hemoglobin and treats it as a fuel gauge. When a BHOC is circulating, that reading combines two populations: the hemoglobin inside cells, and the hemoglobin freely dissolved in plasma. A total hemoglobin of nine may hide a red cell fraction that is far lower.

The hematocrit is even more revealing. It measures the volume fraction of cells and does not include plasma hemoglobin at all. A patient may have a hematocrit that looks alarming, and a total hemoglobin that looks acceptable, and both may be true at once. Clinical trial protocols have had to define separate plasma hemoglobin measurements, often by spinning a sample and reading the clear supernatant, to make sense of this.

The danger is behavioral. A physician who sees a low hematocrit after a BHOC may give a red cell transfusion that was not needed, or refrain from giving one that was. The result is a distorted picture of both safety and efficacy in the data.

Pulse oximetry and the color problem

The finger clip works by shining two wavelengths of light through tissue and comparing absorption. It assumes that the only significant absorbers in arterial blood are oxygenated and deoxygenated hemoglobin. Cell-free carriers add a third pigment and, if the iron oxidizes, a

fourth, methemoglobin. The result is that a saturation reading may be inaccurate in ways that vary by device and by product.

Methemoglobin, the oxidized form that cannot carry oxygen, tends to pull the reading toward the low nineties regardless of the true value. A patient might be badly deprived of oxygen and see a stable number on the monitor, or be adequately oxygenated and see a worrisome one. A carrier that oxidizes quickly, which was a property of some first-generation products, makes this worse.

The laboratory answer is co-oximetry, a bench instrument that measures light absorption at many wavelengths and separates oxyhemoglobin, deoxyhemoglobin, carboxyhemoglobin, and methemoglobin. It is the reference for whether the pigment in the sample is still working. It also has to be calibrated for the specific carrier, since the spectrum of a polymerized bovine hemoglobin is not identical to that of human hemoglobin.

Colorimetric chemistry and the pink plasma

Many routine laboratory tests read a color change. Free hemoglobin in plasma tints the sample and can add to or subtract from the reading. Liver enzyme assays, creatinine methods, bilirubin, and some cardiac markers have shown interference in the presence of free hemoglobin, with the size and direction depending on the method.

This is not an academic footnote. A reported rise in a liver enzyme after a carrier could be genuine injury or an artifact of the assay. A troponin result, used to detect heart muscle damage, is central to the safety debate about hemoglobin carriers, and the reliability of that test in the presence of the product has been discussed in the literature. The way to handle it is to specify in advance which methods are validated for use with the product and to run a second method when the result matters.

Near-infrared spectroscopy: reading the muscle

Near-infrared spectroscopy shines light of several wavelengths through a small volume of tissue and estimates tissue oxygen saturation, usually written StO₂. A common site is the thenar eminence, the fleshy pad at the base of the thumb. The device is noninvasive and can be kept on for hours. It offers something no blood test can: a continuous signal from the microcirculation.

A clever use is the vascular occlusion test. A blood pressure cuff is inflated on the arm until flow stops, tissue saturation falls at a rate that reflects local oxygen consumption, and when the cuff is released the rate of recovery is a reading of microvascular responsiveness. Slow recovery in shock predicts worse outcomes. If a carrier improves recovery, it is doing something at the level that matters.

The limitations are real. Light passes through skin, fat, and muscle, and the signal is a blend of arterial, venous, and capillary blood. Fat thickness changes the reading. Edema, which is common in resuscitation, changes it again. And because the algorithm assumes a certain set of absorbers, a plasma pigment can bias the result. Trend within a patient is more trustworthy than an absolute number, and comparisons between patients or between studies need caution.

Phosphorescence quenching: measuring oxygen by how long light lingers

The most elegant tool in the field is one that most clinicians have never seen. Certain molecules, when excited by light, glow, and the glow fades over a characteristic time. Oxygen quenches that glow. The more oxygen is present, the faster it fades. By measuring the decay time, one can compute the oxygen pressure directly, with no reliance on the color of the blood.

Researchers built probes from palladium porphyrin compounds designed to remain in the circulation or to stay in tissue, and they used them to map oxygen pressure in the microvessels of animals. In experiments, one can look at a single arteriole and a single venule and read the oxygen tension in each. This method is how the field knows, for example, that oxygen leaves the arterioles before it ever reaches the capillaries, a fact that changed how everyone thinks about where a carrier delivers.

It is a research instrument, not a bedside device. It requires a microscope, a probe, and a window into the tissue. But it establishes the physiological truth against which everything else is judged. When a BHOC claims to improve tissue oxygenation, this is the type of measurement that can support or refute the claim.

Functional capillary density: counting the open roads

One of the most influential ideas in this field came from intravital microscopy. In a transparent skin fold chamber on a hamster, capillaries can be watched directly. The count of capillaries with flowing red cells per unit area is called functional capillary density. In hemorrhagic shock it falls, because capillaries collapse or clog, and tissue oxygen falls with it.

The finding that made the field pay attention was that resuscitation with fluids of different properties restored functional capillary density to very different degrees, even when the systemic numbers looked similar. Carriers with high oxygen affinity and low viscosity kept more capillaries open in these experiments, and the proposed reason was that they preserved flow rather than merely adding oxygen content. That idea became the microvascular hypothesis and drove a generation of product design.

The catch is translation. Functional capillary density is measured in a hamster chamber, which cannot be used in a person. The bedside surrogates, sidestream dark-field imaging under the tongue, are useful but variable and depend heavily on operator technique. Whether the hamster result predicts survival in a person remains unresolved.

Lactate and base deficit: the metabolic trail

Lactate is the traditional marker of an oxygen debt. When cells cannot get enough oxygen they switch to a less efficient metabolism that produces lactate, and blood lactate rises. A falling lactate after resuscitation is read as evidence that delivery has improved.

Lactate has flaws. The liver clears it, so a damaged liver can hold it high for reasons unrelated to oxygen. Stress and certain drugs raise it without hypoxia. And it lags: the number describes what has already happened, not what is happening now. Base deficit, a measure of acid load, has similar limits. Both are useful trends. Neither is a direct measurement of tissue oxygenation, and neither is immune to the changes a carrier can make to blood chemistry.

The surrogate trap

Put all of this together and a pattern emerges. The measurements that are easy, such as hemoglobin, saturation, and blood pressure, are the least reliable when a carrier is present. The measurements that are informative, such as phosphorescence quenching and capillary counts, are impossible at the bedside. The middle ground, near-infrared spectroscopy and lactate, is helpful but imperfect.

The regulatory consequence is that trials must decide what to call success. A surrogate endpoint, such as an improved tissue saturation or a fall in lactate, is a stand-in for the outcome that matters, which is survival and recovery. A surrogate is only as good as the evidence that changes in it predict changes in the real outcome. For oxygen carriers, that evidence has been thin, and there is a history of products that improved a surrogate and did not improve survival.

What a careful reader should ask

When a press release or a talk says a carrier improved oxygenation, ask which measurement was used, and whether it was validated in the presence of free hemoglobin. Ask whether the comparison was between patients or within the same patient over time. Ask whether the plasma hemoglobin was reported separately from the red cell hemoglobin. Ask whether the endpoint was a surrogate or a clinical outcome, and if it was a surrogate, what evidence links it to survival.

These are not obstructive questions. They are the ones a trial statistician would ask, and an advocacy organization that can ask them will be listened to.

A modest proposal

The field would benefit from a shared measurement protocol: a defined set of instruments, calibrated for the specific carrier, with a plasma hemoglobin measurement, co-oximetry, a near-infrared trend with the occlusion test, and lactate, all reported in a consistent format. It would not be glamorous. It would let results from different studies be compared, which today they largely cannot.

Until that exists, every claim about a BHOC should be read as a claim about one instrument, on one patient population, in one laboratory. That is not a reason for despair. It is the ordinary discipline of science, and it is what will eventually separate the products that work from those that only look as if they do.

Further reading

Foundational work on phosphorescence quenching oximetry comes from the laboratories of David Wilson and Sergei Vinogradov. The microvascular hypothesis and functional capillary density studies are associated with Marcos Intaglietta and colleagues. Reviews of near-infrared spectroscopy in shock and of interference by cell-free hemoglobin in laboratory assays are available in critical care and clinical chemistry journals. Specific numerical claims here are qualitative and should be checked against those sources.